

Cable Management — Guided Notes ANSWER KEY

Unit 1.3.4 | CompTIA Network+ N10-009 Obj. 1.3 | N10-008 1.3

For instructor use / distribute after students complete the notes.

Question 1

A patch panel is a _____ device (active/passive) that terminates _____ cable runs on its back side using a _____ tool, and provides _____ ports on its front side for connecting to switches.

Answer: Passive; permanent horizontal; 110-style punch-down; RJ-45.

Question 2

The key benefit of using a patch panel is the separation between _____ infrastructure and _____ equipment. To move a user to a different switch port, you only need to move the _____ — the cable in the wall never changes.

Answer: Permanent (horizontal) infrastructure; active switching equipment; patch cable on the front of the panel.

Question 3

_____ cable managers mount between devices in a rack to route patch cables side to side. _____ cable managers run along the sides of the rack to contain bulk cables from top to bottom. _____ are open or enclosed metal troughs in ceilings or raised floors that support horizontal cable runs.

Answer: Horizontal; vertical; cable trays.

Question 4

Explain why zip ties should NOT be used on data cable bundles, and what should be used instead.

Answer: Over-tightened zip ties deform the cable jacket and change the geometry of the twisted pairs inside, causing crosstalk and attenuation that may not appear immediately but develops over time.

Velcro (hook-and-loop) straps should be used instead — they hold firmly without crushing the cable and can be reopened when cables are added or removed.

Question 5

When installing a cable drop to Room 214, outlet A, what three things get labeled '214-A'?

1. _____
2. _____
3. _____

Answer: 1. The wall jack in Room 214; 2. The corresponding patch panel port; 3. Both ends of the cable run in the ceiling.

Question 6

One rack unit (1U) equals _____ inches. A standard full-height rack is _____ U. Heavy equipment such as _____ should be mounted at the _____ of the rack for stability. Racks should be left _____ % empty for future expansion.

Answer: 1.75 inches; 42U; UPS units; bottom; 20-30%.

Question 7

The _____ standard defines labeling conventions for commercial cabling systems. Network documentation should be stored _____ (location), not in one person's head, and must include at minimum: _____, _____, and _____.

Answer: TIA-606; in a shared location accessible to the whole team (wiki, NMS, shared drive); physical layer diagram, logical diagram with VLAN/switch port assignments, and a change log.

Question 8 — Real World Application

REAL WORLD: You're handed a network closet that has no labels, cables draped over equipment, and no documentation. You need to find which patch panel port connects to a specific classroom without taking the network down. Describe your approach step by step.

Answer: 1. Get a cable toner/tracer — tone the wall jack in the classroom and trace the tone to the patch panel to identify the port. 2. Label that port immediately. 3. Check which switch port it patches to and document that mapping. 4. Repeat for all active ports, building a port-to-room map as you go. 5. Once all active ports are mapped, create printed labels for the panel and a physical diagram. Document in a shared location before leaving the closet.